

Question number	Answer	Mark
1(a)	B inner mitochondrial membrane	(1)

Question number	Answer	Mark
1(b)(i)	An explanation that includes any two of the following points: • no ATP will be produced by the electron transport chain (1) • because electrons will not be passed onto the oxygen (1) • therefore {reduced NAD will not be reoxidised / the electron transport chain will stop} (1)	(2)

Question number	Answer	Mark
1(b)(ii)	An explanation that includes the following points: • there will be no ATP production from reduced NAD (1) • there will be some ATP produced from reduced FAD because electrons from reduced FAD enter the electron transport chain after the site of inhibition of rotenone (1)	(2)

Question number	Answer	Mark
1(b)(iii)	An explanation that includes the following points: • because the inhibitors affect the ETC, which is not used in anaerobic respiration (1) • therefore the same quantity of ATP will be produced from substrate	(2)